

OBSERVATION TASK 1.2

Course :	Full Stack Web Development
Course Code:	23CM4121
Observation Task No.:	1.2
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1. TITLE

My Styled Webpage using HTML5 Canvas and CSS3

2. AIM

To design and develop a visually appealing webpage titled **“My Styled Webpage”** using **HTML5 Canvas and CSS3**, demonstrating basic graphics such as a rectangle, circle, line, and text, along with CSS3 styling features including colors, fonts, borders, gradients, spacing, shadows, rounded corners, and hover effects.

3. OBJECTIVE

- To understand and implement the HTML5 Canvas element for drawing basic graphics.
- To create a webpage using CSS syntax and selectors.
- To apply CSS3 properties such as colors, fonts, borders, margins, padding, and backgrounds.
- To implement gradients, rounded corners, box shadows, and text shadows.
- To create a structured layout with a header, navigation bar, three content cards, and footer.
- To apply hover effects to navigation links and buttons for better user interaction.
- To develop a visually attractive and well-structured webpage using HTML5 and CSS3.

4. THEORY

HTML5 Canvas is an HTML element used to draw graphics dynamically on a webpage using JavaScript. It provides a drawing area where shapes such as rectangles, circles, and lines can be created. Text can also be displayed on the canvas using JavaScript drawing methods.

CSS3 (Cascading Style Sheets) is used to control the appearance and layout of HTML elements. CSS uses selectors and properties to apply styles such as colors, fonts, borders, margins, padding, backgrounds, and shadows.

CSS3 provides advanced styling features such as gradients, rounded corners, box shadows, text shadows, and hover effects. These properties help create modern and visually attractive webpages. Hover effects allow webpage elements such as navigation links and buttons to change their appearance when the user moves the mouse over them.

In this task, HTML5 Canvas is used to create basic graphics, while CSS3 is used to design the header, navigation bar, content cards, buttons, and footer. Together, HTML5 and CSS3 provide a structured, interactive, and visually appealing webpage.

5. ALGORITHM / PROCEDURE

1. Create a basic **HTML5 document structure** using `<!DOCTYPE html>`.
 2. Add the webpage title **"My Styled Webpage"**.
 3. Create a **header** containing the webpage heading.
 4. Add a **navigation bar** with links to different sections.
 5. Create an HTML5 **<canvas>** element for drawing graphics.
 6. Use JavaScript with the Canvas API to draw a **rectangle, circle, line, and name as text**.
 7. Create three **content cards** to display webpage information.
 8. Create a **footer** for the webpage.
 9. Apply CSS3 selectors and properties for **colors, fonts, borders, margins, and padding**.
 10. Apply **gradient backgrounds, rounded corners, box shadows, and text shadows** to improve the appearance.
 11. Add **hover effects** to navigation links and buttons.
 12. Save the HTML, CSS, and JavaScript files and open the webpage in a browser.
 13. Verify that all Canvas graphics and CSS3 styling features are displayed correctly.
-

6. PROGRAM

A : Index.html

```
<!DOCTYPE html>  
<html lang="en">
```

```

<head>
  <!-- Defines character encoding -->
  <meta charset="UTF-8">

  <!-- Makes the webpage responsive -->
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>My Styled Webpage</title>

  <!-- Linking external CSS file -->
  <link rel="stylesheet" href="styleobtask1.2.css">
</head>

<body>

  <!-- ===== HEADER ===== -->
  <header>
    <h1>My Styled Webpage</h1>
    <p>HTML5 Graphics and CSS3 Web Styling</p>
  </header>

  <!-- ===== NAVIGATION ===== -->
  <nav>
    <a href="#home">Home</a>
    <a href="#canvas">Canvas</a>
    <a href="#cards">Cards</a>
    <a href="#contact">Contact</a>
  </nav>

  <!-- ===== CANVAS SECTION ===== -->
  <section id="canvas">

    <h2>HTML5 Canvas</h2>

    <!-- Canvas drawing area -->
    <canvas id="myCanvas"
      width="600"
      height="300">
      Your browser does not support HTML5 Canvas.
    </canvas>

  </section>

  <!-- ===== CONTENT CARDS ===== -->
  <section id="cards">

    <h2>Our Features</h2>

    <div class="card-container">

```

```
<!-- Card 1 -->
<div class="card">

  <h3>HTML5</h3>

  <p>
    HTML5 provides modern elements for creating
    structured and interactive webpages.
  </p>

  <button>Learn More</button>

</div>

<!-- Card 2 -->
<div class="card">

  <h3>CSS3</h3>

  <p>
    CSS3 is used to design webpages using colors,
    fonts, borders, shadows and layouts.
  </p>

  <button>Learn More</button>

</div>

<!-- Card 3 -->
<div class="card">

  <h3>Graphics</h3>

  <p>
    HTML5 Canvas allows developers to create
    shapes, lines and text dynamically.
  </p>

  <button>Learn More</button>

</div>

</div>

</section>

<!-- ===== CONTACT SECTION ===== -->
<section id="contact">

  <h2>Contact</h2>
```

```
<p>
  Thank you for visiting My Styled Webpage.
</p>

</section>

<!-- ===== FOOTER ===== -->
<footer>

  <p>
    &copy; 2026 My Styled Webpage | HTML5 & CSS3
  </p>

</footer>

<!-- ===== CANVAS JAVASCRIPT ===== -->
<script>

  // Get the Canvas element
  const canvas = document.getElementById("myCanvas");

  // Get the 2D drawing context
  const ctx = canvas.getContext("2d");

  // ----- Draw Rectangle -----

  ctx.fillStyle = "lightblue";

  ctx.fillRect(50, 50, 150, 80);

  // ----- Draw Circle -----

  ctx.beginPath();

  ctx.arc(320, 90, 45, 0, 2 * Math.PI);

  ctx.fillStyle = "lightgreen";

  ctx.fill();

  ctx.stroke();

  // ----- Draw Line -----

  ctx.beginPath();

  ctx.moveTo(400, 50);

  ctx.lineTo(520, 130);

  ctx.strokeStyle = "black";
```

```

    ctx.lineWidth = 3;

    ctx.stroke();

    // ----- Display Name -----

    ctx.font = "24px Arial";

    ctx.fillStyle = "black";

    ctx.fillText("R. Srivalli", 210, 230);

</script>

</body>

</html>

```

B.STYLE.CSS

```

/* ===== GENERAL STYLING ===== */

* {
  box-sizing: border-box;
}

body {
  margin: 0;
  font-family: Arial, sans-serif;
  background-color: #f4f4f4;
  color: #222;
}

/* ===== HEADER ===== */

header {
  padding: 30px;
  text-align: center;

  /* Gradient background */
  background: linear-gradient(135deg, #4facfe, #6a5acd);

  color: white;

  /* Text shadow */
  text-shadow: 2px 2px 4px #333;
}

header h1 {

```

```

margin: 0;
font-size: 36px;
}

header p {
font-family: Georgia, serif;
font-size: 18px;
}

/* ===== NAVIGATION ===== */

nav {
background-color: #222;
padding: 15px;
text-align: center;
}

nav a {
color: white;
text-decoration: none;
font-size: 17px;
margin: 0 15px;
padding: 8px 12px;

/* Rounded corners */
border-radius: 6px;
}

/* Navigation hover effect */

nav a:hover {
background-color: #4facfe;
color: black;
}

/* ===== SECTIONS ===== */

section {
padding: 30px;
margin: 20px auto;
max-width: 1000px;
}

section h2 {
text-align: center;
font-family: Georgia, serif;
font-size: 28px;

text-shadow: 1px 1px 2px #aaa;
}

```

```
/* ===== CANVAS ===== */

canvas {
  display: block;
  margin: 20px auto;

  background-color: white;

  border: 3px solid #333;

  border-radius: 12px;

  /* Box shadow */
  box-shadow: 0 5px 15px #999;
}

/* ===== CARD CONTAINER ===== */

.card-container {
  display: flex;
  justify-content: center;
  gap: 25px;
  flex-wrap: wrap;
}

/* ===== CARDS ===== */

.card {
  width: 280px;

  background-color: white;

  border: 2px solid #ddd;

  border-radius: 15px;

  padding: 25px;

  margin: 10px;

  box-shadow: 0 5px 15px #bbb;

  text-align: center;

  transition: 0.3s;
}

.card h3 {
  font-family: Georgia, serif;
  font-size: 24px;
}
```

```

.card p {
  font-size: 16px;
  line-height: 1.6;
}

/* Card hover effect */

.card:hover {
  transform: translateY(-8px);

  box-shadow: 0 10px 25px #888;
}

/* ===== BUTTON ===== */

button {
  background-color: #4facfe;

  color: white;

  border: none;

  border-radius: 8px;

  padding: 10px 20px;

  font-size: 15px;

  cursor: pointer;

  transition: 0.3s;
}

/* Button hover effect */

button:hover {
  background-color: #222;

  color: white;

  transform: scale(1.05);
}

/* ===== CONTACT ===== */

#contact {
  text-align: center;

  background: linear-gradient(
    135deg,

```

```
    #fffff,  
    #e8f4ff  
);  
  
border: 1px solid #ddd;  
  
border-radius: 12px;  
}  
  
/* ===== FOOTER ===== */  
  
footer {  
    margin-top: 30px;  
  
    padding: 20px;  
  
    background-color: #222;  
  
    color: white;  
  
    text-align: center;  
  
    font-size: 14px;  
}  
  
/* ===== RESPONSIVE DESIGN ===== */  
  
@media (max-width: 700px) {  
  
    header h1 {  
        font-size: 28px;  
    }  
  
    nav a {  
        display: inline-block;  
        margin: 5px;  
    }  
  
    canvas {  
        width: 100%;  
        height: auto;  
    }  
  
    .card {  
        width: 90%;  
    }  
}
```

7. CODE EXPLANATION

S.No.	HTML/CSS/JavaScript Component	Explanation
1	<!DOCTYPE html>	Declares that the document uses HTML5.
2	<html>	Defines the root element of the webpage.
3	<head>	Contains metadata, title, and links to external resources.
4	<title>	Sets the title displayed on the browser tab.
5	<link rel="stylesheet">	Connects the external CSS file with the HTML document.
6	<header>	Creates the top section containing the webpage heading.
7	<nav>	Creates the navigation bar with links to different sections.
8	<main>	Contains the main content of the webpage.
9	<canvas>	Provides a drawing area for creating graphics using JavaScript.
10	getContext("2d")	Obtains the 2D drawing context required for Canvas operations.
11	fillRect()	Draws and fills a rectangle on the Canvas.
12	arc()	Creates a circular or curved path on the Canvas.
13	moveTo() and lineTo()	Define the starting and ending po

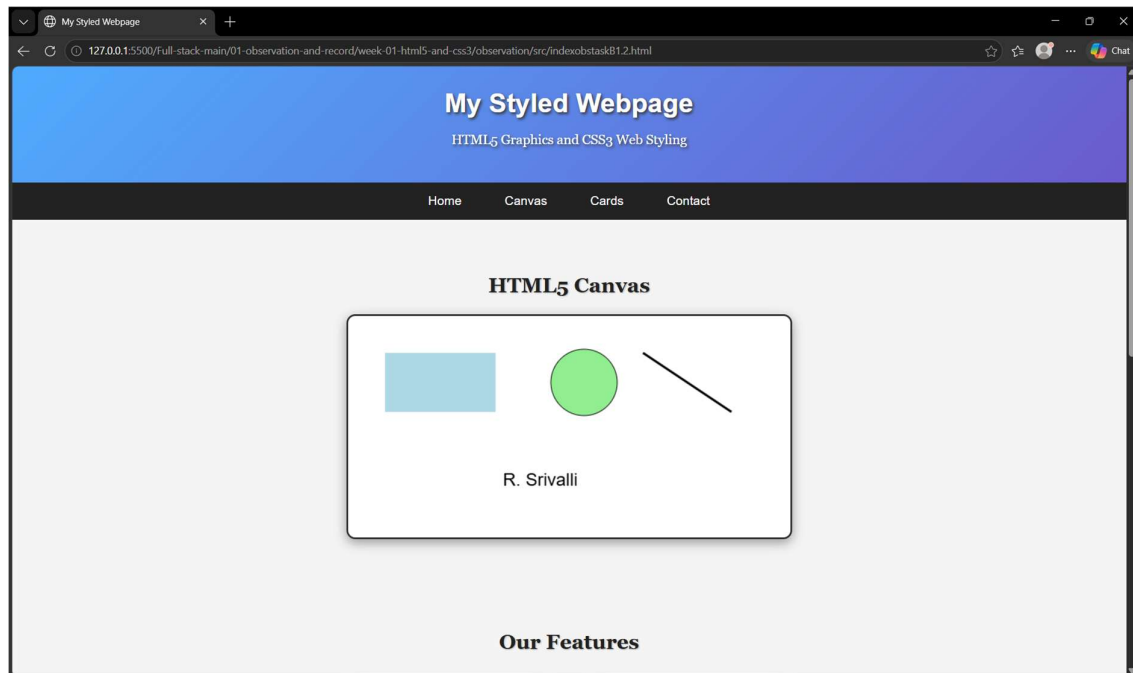
8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Status
TC01	Open the HTML file in a browser	"My Styled Webpage" loads successfully	Pass

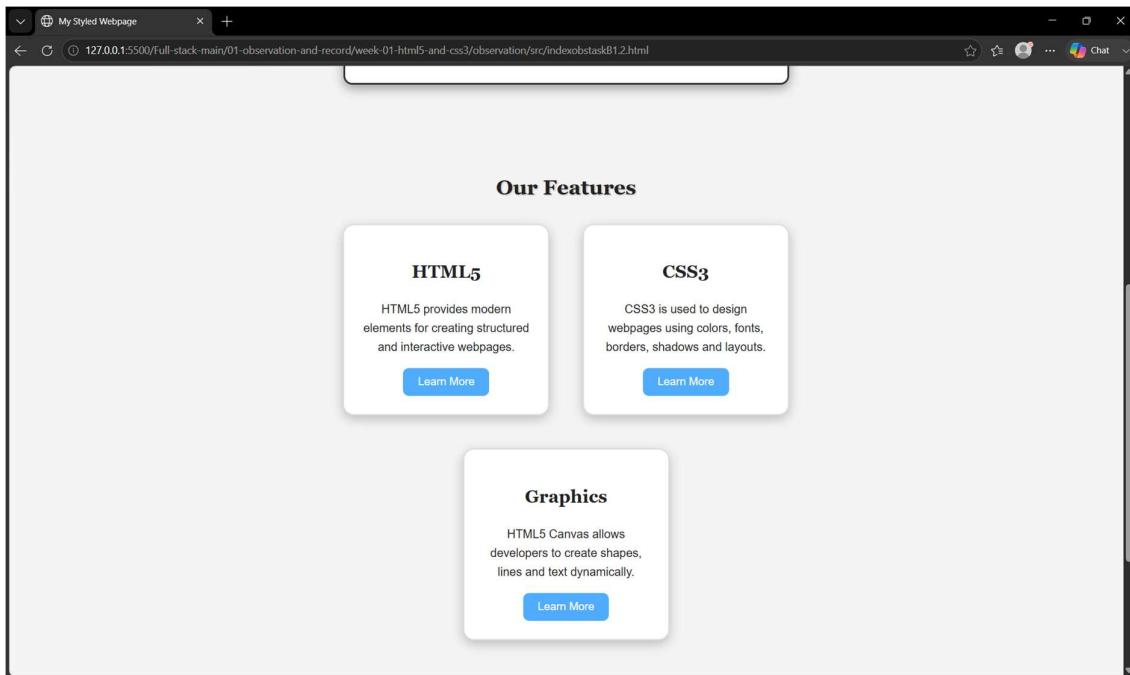
TC02	View the Canvas section	Rectangle, circle, line, and name are displayed	Pass
TC03	Check the header and navigation bar	Header and navigation links are displayed correctly	Pass
TC04	View the three content cards	All three cards are displayed with proper styling	Pass
TC05	Move the pointer over navigation links	Navigation links change appearance on hover	Pass
TC06	Move the pointer over buttons	Button hover effect is applied	Pass
TC07	Check page styling	Colors, fonts, borders, gradients, spacing, rounded corners, and shadows are applied correctly	Pass
TC08	Scroll to the bottom of the webpage	Footer is displayed correctly	Pass

9. OUTPUT

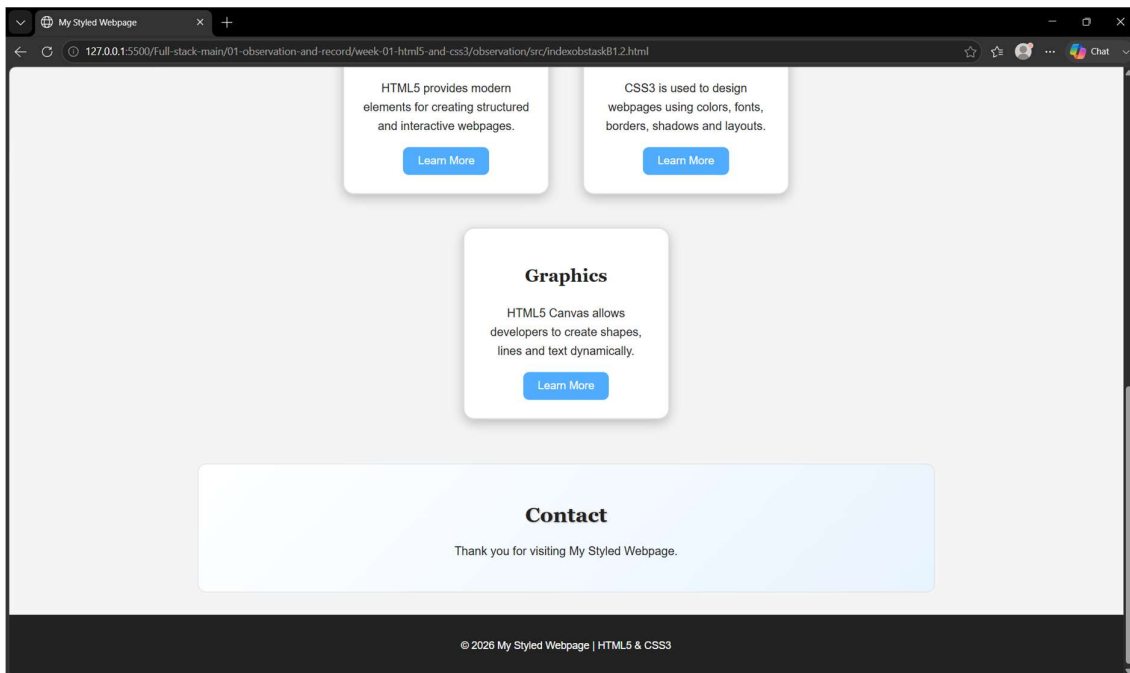
Output 1:



Output 2:



Output3:



10. RESULT

The “My Styled Webpage” was successfully designed and developed using HTML5 Canvas and CSS3. The webpage demonstrates basic Canvas graphics such as a rectangle, circle, line, and text, along with CSS3 features including colors, fonts, borders, gradients, spacing, shadows, rounded corners, and hover effects. All major components were tested and displayed successfully.